INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

September 2025

Subject CS1 – Actuarial Statistics
Core Principles

Paper A

Time allowed: Three hours and twenty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

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1 Consider discrete random variables X and Y, with joint probability function given by:

$$p_{X,Y}(x,y) = \frac{e^{-y}y^x}{2^y x!}, x = 0,1,2,\ldots;\ y = 1,2,3,\ldots$$

The marginal distribution of Y is Geometric(p) with probability function $p(y) = p\,(1-p)^{y-1}$, with $p = 0.5$. (You are not required to check this result.)

(i) Identify which one of the following options gives the correct expression for the conditional probability function $p_{X|Y=y}(x|y)$:

A $\frac{e^{-2y}(2y)^x}{x!}$

B $\frac{e^{-y}y^x}{x!}$

C $\frac{e^{-\frac{y}{2}}y^x}{2^x x!}$

D $\frac{e^{-y^2}y^{2x}}{x!}$.

[2]

(ii) State the distribution of X conditional on $Y = y$, and its parameter. [1]

(iii) Calculate the expected value of X. [2]

[Total 5]

2 A random sample of size $n = 50$ is taken from a normal distribution with unknown expectation μ and known variance $\sigma^2 = 16$. The sample mean is used to estimate the parameter μ.

(i) Determine if the sample mean is an unbiased estimator of μ justifying your answer. [2]

(ii) Calculate the variance of the sample mean. [2]

(iii) Determine the mean square error of the sample mean. [1]

[Total 5]

3 A random variable, X, has a moment generating function $M_X(t)$ given by:

$$M_X(t) = (1 + a)^{t^2+1}$$

where $a > 0$ is a fixed parameter.

(i) Determine the cumulant generating function, $C_X(t)$. [2]

(ii) Determine, using your answer to part (i), the mean and variance of X in terms of a. [3]

[Total 5]

4 An actuary is analysing the variance in monthly claim amounts (in units of £1,000) reported in two insurance portfolios, Portfolio A and Portfolio B. The actuary collects two independent random samples:

- Sample A: a sample of 25 months from Portfolio A with a sample variance of 25.
- Sample B: a sample of 13 months from Portfolio B with a sample variance of 15.

The actuary wants to determine if the variances of the claim amounts in the two portfolios, σ_A^2 and σ_B^2, are significantly different.

(i) Determine a 90% confidence interval for the ratio of the population variances $\frac{\sigma_A^2}{\sigma_B^2}$. [4]

(ii) Test the hypothesis $\frac{\sigma_A^2}{\sigma_B^2} = 1$ versus $\frac{\sigma_A^2}{\sigma_B^2} \neq 1$ at the 10% level of significance using your answer to part (i). [2]

(iii) State the two assumptions required for the confidence interval in part (i) and the test in part (ii) to be appropriate. [1]

[Total 7]

5 An insurance company wishes to use the normal/normal Bayesian model to estimate the expected aggregate claims cost for a particular risk for the coming year.

Let X_i be random variables representing the aggregate claims cost for years $i = 1, 2, 3, \ldots$.

The distribution of each X_i depends on the fixed, but unknown, value of a parameter, θ.

The conditional distribution of X_i given θ is $N(\theta, \sigma_1^2)$. Given θ, the random variables $\{X_i\}$ are independent.

The prior distribution of θ is $N(\mu, \sigma_2^2)$.

(i) Discuss if the normal/normal Bayesian model, including its parameters, is suitable for modelling aggregate claims cost. [2]

(ii) Show that for all i, $\text{Var}[X_i] = \sigma_1^2 + \sigma_2^2$. [2]

[**Hint:** Use the result involving the conditional expectation and variance from *Formulae and Tables for Examinations of The Institute and Faculty of Actuaries*.]

Let $\rho_{i,j}$ be the (unconditional) correlation coefficient between X_i and X_j.

(iii) Identify which one of the following options gives the correct expression for $\rho_{i,j}$, where $i \neq j$:

A $\rho_{i,j} = \dfrac{\sigma_2^2}{\sqrt{\sigma_1^2 + \sigma_2^2}}$

B $\rho_{i,j} = \dfrac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$

C $\rho_{i,j} = \dfrac{\sigma_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}$

D $\rho_{i,j} = \dfrac{\sigma_2^2}{(\sigma_1 + \sigma_2)^2}$.

[3]

(iv) State the value of $\rho_{i,j}$ for each of the following extreme cases and explain the validity of your answers in the context of the normal/normal Bayesian model:

(a) $\sigma_1^2 = 0$

(b) $\sigma_2^2 = 0$.

[4]

[Total 11]

6 An insurance company has recently launched a policy covering a specific medical condition. It collected data for the first 100 policyholders and found that 15 of them made claims in the first year of coverage. It is assumed that policyholders make claims independently from one another.

The company wants to estimate the probability, θ, of a policyholder making a claim for the condition in the first year of coverage. It uses Bayesian inference with a prior belief that θ follows a Beta(a, b) distribution, with parameters $a > 0$ and $b > 0$.

(i) Determine the posterior distribution of θ, including its parameters, in terms of a and b. [5]

(ii) Comment on the relationship between the prior and the posterior distribution of θ using your answer in part (i). [1]

(iii) Show that the Bayesian estimate of parameter θ under all-or-nothing loss is:

$$\theta' = \frac{a+14}{a+b+98}.$$

[3]

(iv) Calculate θ' for a prior distribution Beta(1, 1). [1]

The company has also calculated the Maximum Likelihood Estimate (MLE) of θ as $\hat{\theta} = 0.15$.

(v) Comment on the comparison of the MLE and your answer in part (iv). [2]

[Total 12]

7 A chain of fast-food restaurants is experiencing issues with their burgers, with some customers suffering from food poisoning. A sample of five customers has been surveyed in each of the 100 restaurants in this chain.

The table below shows the number of sampled customers, x, who have experienced food poisoning in the past month and the corresponding frequency, f, in terms of the number of restaurants.

Number of customers with food poisoning, x	0	1	2	3	4	5
Number of restaurants, f	40	35	19	4	1	1

Assume that for each restaurant, the number of sampled customers that have suffered from food poisoning during the last month follows a binomial distribution, i.e.:

$$X \sim \text{Bin}(5, p)$$

where p is the probability that a customer has suffered food poisoning during the last month.

(i) State the assumptions needed to justify the use of the binomial distribution for X. [1]

(ii) Identify which one of the following gives the likelihood function for the parameter, p, based on the above data. For all options, c denotes a constant that does not depend on the parameter p.

A $L(p) = c \times p^{94} \times (1 - p)^{406}$

B $L(p) = c \times p^{90} \times (1 - p)^{410}$

C $L(p) = c \times p^{406} \times (1 - p)^{94}$

D $L(p) = c \times p^{410} \times (1 - p)^{90}$.

[3]

(iii) Derive the maximum likelihood estimate of p and verify that the obtained estimate maximises the likelihood function. [4]

(iv) Determine, for each value of x, the corresponding expected number of restaurants based on the assumed binomial model and the estimate found in part (iii). [3]

Each food poisoning incident costs the restaurant money in refunds and compensation. Data collected for 30 food poisoning incidents reveals:

$$\sum_{i=1}^{30} y_i = 975\,, \quad \sum_{i=1}^{30} {y_i}^2 = 34062$$

where y_1, …, y_{30} denote the costs per incident for each of the 30 incidents.

(v) Calculate an approximate 95% confidence interval for the mean cost per incident, stating any necessary assumptions. [5]

[Total 16]

8 A quality control inspector at an electronics manufacturing plant checks electronic components coming off an assembly line. For each component, there is a 7% chance of it being defective, independent of other components.

Let X represent the number of components required to be checked to find the first defective component. X will therefore have a geometric distribution with probability function $p(x) = p\,(1-p)^{x-1}$, $x = 1, 2, \ldots$, for a suitable parameter p.

(i) Calculate the expected value of X. [2]

(ii) Calculate the probability that the inspector does not find a defective component in the first four checks. [2]

(iii) Describe what is meant by the 'memoryless' property of the geometric distribution. [2]

(iv) Calculate the probability that the inspector does not find a defective component in the first ten checks, given that a defect has not been found in the first four checks. [2]

The managers of the manufacturing plant decide to implement a new automated quality control system involving two robotic scanners. If a scanner does not detect a defect when scanning a component, it is said to 'pass' the component, otherwise it 'fails' the component.

Scanner A checks components first. It has a 75% chance of correctly identifying a defective component and a 95% chance of correctly identifying a non-defective component.

If a component is passed by Scanner A, it moves on to Scanner B. Scanner B has an 85% chance of correctly identifying a defective component and a 90% chance of correctly identifying a non-defective component, independently of Scanner A.

A component is accepted if it is passed by both scanners.

(v) Calculate the probability that a component is accepted (passed by both scanners) when:

(a) the component is defective.

(b) the component is non-defective.

[3]

(vi) Identify which one of the following options gives the correct probability that an accepted component is actually defective:

A 0.263%

B 0.329%

C 0.525%

D 0.658%.

[3]

On a particular day, the automated system of both scanners has scanned a number of components, of which 4,512 were accepted.

Each incorrectly accepted component results in a liability cost to the manufacturing plant of $200.

(vii) Calculate the expected total liability cost of the incorrectly accepted components. [1]

(viii) Calculate, using a suitable approximation, the probability that the total liability cost of the incorrectly accepted components will exceed $5,000. [4]

[Total 19]

9 An actuary at a health insurer is initiating an investigation to model its claim costs for critical illness cover using a simple generalised linear model with the following factors used:

- NWL Net Wealth Level
- SMK Smoker Status
- CRR Main Residency.

The factors are defined as follows:

$$\text{NWL} \in \{1, 2, 3, 4, 5\}$$

where 1 is lowest net wealth level and 5 is the highest net wealth level.

$$\text{SMK} = \begin{cases} 0, & \text{Non-smoker} \\ 1, & \text{Smoker} \end{cases}$$

$$\text{CRR} = \begin{cases} 0, & \text{City main residency} \\ 1, & \text{Rural main residency} \end{cases}$$

The actuary is considering four potential models for the linear predictor as shown below:

Model 1: NWL + SMK
Model 2: NWL + SMK + CRR
Model 3: NWL + SMK + NWL:SMK + CRR
Model 4: NWL*SMK*CRR

where:

NWL:SMK denotes the interaction term between these respective variables.
NWL*SMK*CRR denotes the main effects and the interaction terms between these respective variables.

(i) Determine the number of parameters in each of the four models, justifying your answer. [4]

(ii) Discuss the suitability of models 1, 2 and 4 for modelling claim costs, in terms of inclusion of the three variables NWL, SMK and CRR. [3]

The actuary has produced the summary table below, which shows the values of the scaled deviances for these four models including the constant model (denoted as Model 0).

Model number	*Model*	*Scaled deviance*	*Degrees of freedom*
0		86.5	19
1	NWL + SMK	68.6	
2	NWL + SMK + CRR	45.7	
3	NWL + SMK + NWL:SMK + CRR	11.3	
4	NWL*SMK*CRR	0	

(iii) Calculate the missing entries for the degrees of freedom column. [2]

(iv) Determine, by sequentially comparing each model to the next one (starting with Model 0), which model should be chosen based on differences of the scaled deviances at the 5% level. [9]

(v) Outline extra information the actuary could consider before any of the four models are chosen. [2]

[Total 20]

END OF PAPER